PRIME PALINDROME NUMBER CHECK

Java Practical Record

1. Aim

To write and execute a Java program that checks whether a given integer is both a prime number and a palindrome number.

2. Description

The program reads an integer from the user. The **checkPrime()** method checks whether the number is prime by testing divisors from 2 up to the square root of the number. The **checkPalindrome()** method reverses the digits and compares the reversed number with the original. If both methods return true, the number is reported as a Prime Palindrome number; otherwise, it is reported as not a Prime Palindrome number.

3. System Requirements

• Hardware: Computer or laptop with a keyboard and monitor.
• Processor: Any processor capable of running a Java development environment.
• Memory: 2 GB RAM or more recommended.
• Software: Java Development Kit (JDK) and a text editor or IDE such as Notepad, VS Code, Eclipse, or IntelliJ IDEA.
• Operating System: Windows, Linux, or macOS.

4. Algorithm

1. Start the program.

2. Create a Scanner object and read an integer from the user.

3. Call checkPrime(value) to test whether the number is prime.

4. In checkPrime(), return false if the number is less than 2.

5. Check divisors from 2 while i × i is less than or equal to the number. If any divisor divides it exactly, return false; otherwise return true.

6. Call checkPalindrome(value). Store the original number and reverse its digits using remainder (%) and division (/).

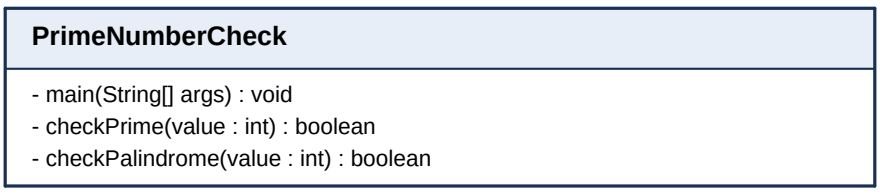
7. Compare the original number with the reversed number.

8. If both checks are true, print that it is a Prime Palindrome number.

9. Otherwise, print that it is not a Prime Palindrome number.

10. Close the Scanner and stop.

5. UML Class Diagram



The main() method uses checkPrime() and checkPalindrome() to decide the final result.

6. Program Code

```
package numberprog;

import java.util.Scanner;

public class PrimeNumberCheck {

    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        System.out.println("Enter a number");
        int value = input.nextInt();

        boolean prime = checkPrime(value);
        boolean palindrome = checkPalindrome(value);

        if (prime && palindrome) {
            System.out.println(value + " is a Prime Palindrome number");
        } else {
            System.out.println(value + " is not a Prime Palindrome number");
        }

        input.close();
    }

    // Method to check prime number
    static boolean checkPrime(int value) {
        if (value < 2)
            return false;

        for (int i = 2; i * i <= value; i++) {
            if (value % i == 0)
                return false;
        }

        return true;
    }

    // Method to check palindrome number
    static boolean checkPalindrome(int value) {
        int original = value;
        int reverse = 0;

        while (value != 0) {
            int digit = value % 10;
            reverse = reverse * 10 + digit;
            value = value / 10;
        }

        if (original == reverse)
            return true;
        else
            return false;
    }
}
```

7. Output Screenshot

Sample run for input 131 (131 is prime and its digits read the same forward and backward):

```
Enter a number
131 is a Prime Palindrome number
```

Additional sample output (input 12)

```
Enter a number
12 is not a Prime Palindrome number
```

8. Result

Thus, the Java program was written and executed successfully. It checks whether the entered number is both prime and palindrome and displays the appropriate message.